IBEROAMERICAN JOURNAL OF MEDICINE
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Journal homepage: www.iberoamjmed.com

Original article

Determining factors in angular deformity of the knee: influence of age, gender and BMI. A preoperative study

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ARTICLE INFO

Article history:
Received 13 May 2026
Received in revised form 20 August 2026
Accepted 13 September 2026

Keywords:
Knee
Angular deformity
Varus
Valgus
Total knee replacement

ABSTRACT

Introduction: Knowing and properly assessing the anthropometric characteristics of patients and the type and severity of femorotibial deformity can allow for planning and executing a better surgical technique in total knee replacement.
Material and methods: A descriptive and correlational study of 130 patients was conducted. The angular alignment of the leg axis was analysed using a long-leg radiograph that included the hip, knee, and ankle. 11.5% of cases were classified as having a neutral axis, 26.1% as valgus, and 62.3% as varus. The absolute value of the deformity was used for severity analysis.
Results: A positive correlation was observed between age and the severity of the deformity ($r = 0.28$, $p = 0.001$). The varus pattern was predominant in men (77.8%) and showed the strongest association with age ($r = 0.42$, $p < 0.001$), while valgus was more frequent in women. Patients with varus had a significantly higher BMI than those with valgus ($p < 0.001$). When severe deformities, greater than 20°, were present, a significantly higher BMI was also observed compared to the other cases ($p < 0.05$).
Conclusions: Older age and higher BMI are critical predictors of the severity of angular deformity, especially in the varus pattern. There is a marked sexual dimorphism, with men showing a linear progression toward varus and women accounting for most severe valgus cases. These findings underscore the importance of weight management and close follow-up in elderly patients to prevent severe joint collapse and technical difficulties in total knee replacement.

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https://doi.org/10.53986/ibjm.2026.0024

Factores determinantes de la deformidad angular de la rodilla: influencia de la edad, el sexo y el IMC. Un studio preoperatorio

INFO. ARTÍCULO

Historia del artículo:
Recibido 13 Mayo 2026
Recibido en forma revisada 20 Agosto 2026
Aceptado 13 Septiembre 2026

Palabras clave:
Rodilla
Deformidad angular
Varo
Valgo
Artroplastia total de rodilla

RESUMEN

Introducción: Conocer y evaluar adecuadamente las características antropométricas de los pacientes, así como el tipo y la gravedad de la deformidad femorotibial, permite planificar y ejecutar una mejor técnica quirúrgica en la artroplastia total de rodilla.
Material y métodos: Se realizó un estudio descriptivo y correlacional con 130 pacientes. Se analizó la alineación angular del eje de la extremidad inferior mediante una radiografía de miembro completo que incluía cadera, rodilla y tobillo. El 11,5 % de los casos presentó un eje neutro; el 26,1 %, valgo; y el 62,3 %, varo. Para analizar la gravedad, se utilizó el valor absoluto de la deformidad.
Resultados: Se observó una correlación positiva entre la edad y la gravedad de la deformidad ($r = 0,28$; $p = 0,001$). El patrón en varo predominó en los hombres (77,8 %) y mostró la asociación más fuerte con la edad ($r = 0,42$; $p < 0,001$), mientras que el valgo fue más frecuente en las mujeres. Los pacientes con varo presentaron un IMC significativamente mayor que aquellos con valgo ($p < 0,001$). Asimismo, en los casos de deformidades graves (superiores a 20°), se observó un IMC significativamente más elevado en comparación con el resto de los casos ($p < 0,05$).
Conclusiones: La edad avanzada y un IMC elevado son factores predictores clave de la gravedad de la deformidad angular, especialmente en el patrón en varo. Existe un marcado dimorfismo sexual: los hombres muestran una progresión lineal hacia el varo, mientras que las mujeres representan la mayoría de los casos graves de valgo. Estos hallazgos subrayan la importancia del control del peso y de un seguimiento estrecho en pacientes de edad avanzada para prevenir el colapso articular grave y las dificultades técnicas durante la artroplastia total de rodilla.

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HOW TO CITE THIS ARTICLE: Hernandez-Vaquero D, Noriega-Fernandez A, Iglesias-Fernandez S. Determining factors in angular deformity of the knee: influence of age, gender and BMI. A preoperative study. Iberoam J Med. 2026;8(4):133-139. doi: 10.53986/ibjm.2026.0024.

1. INTRODUCTION

Understanding the mechanical alignment of the leg axis is essential for performing a total knee replacement (TKR) correctly, and therefore a long-leg radiograph or CT is mandatory for designing the surgical technique. Combining interventions on the bone and soft tissues should help achieve correct leg alignment after the TKR. However, axis alignment, assessed in isolation, does not adequately inform about the actual situation of the knee, and other factors can modify the surgical approach. Anthropometric factors associated with leg axis deformities in the coronal plane have been studied in the literature, but few studies analyse them in the immediate preoperative period. If the patient is not properly positioned at the time of the TKR, the outcome of this procedure cannot be fairly analysed. It is possible that some published studies on the evolution of TKRs do not show the true initial clinical and radiographic situation of the limb, and therefore their results may be poorly estimated [1]. Knowing and properly assessing the anthropometric characteristics of patients also allows for planning and carrying out a better surgical technique.

Our work begins with the examination of long-leg radiographs of the limb taken from a group of patients immediately before TKR. We analysed the type and severity of femorotibial deformity in the coronal plane and looked for its possible correlation with the anthropometric data of the series: age, gender, and BMI.

2. MATERIAL AND METHODS

All the procedures performed in this study were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. The study was approved by the Ethical Committee. Informed consent was obtained from all patients. A descriptive and correlational study was conducted on a

sample of 130 cases that were studied prior to TKR with a navigation system. Forty-five cases were male and 85 were female. Patients were consecutively enrolled in the study upon being placed on the surgical waiting list. A full-lengh radiograph of the limb, including the hip, knee, and ankle, was obtained in all cases. Using specific software (Impax 6.3.1.2813, Agfa Healthcare N.U. Montsel, Belgium), the images were sent to the surgical planning software (Agfa Orthopaedics Tools version 2.06). This tool was used to first calculate the mechanical axes of the femur and tibia, and then the mechanical axis of the entire leg. To assess lower limb angular deformity, the long-leg radiograph (LRx), measured in degrees, was used, categorizing the results into three groups according to alignment: neutral range (between -3º and +3º), valgus (< -3º), and varus (> +3º). Additionally, demographic and anthropometric variables such as age, sex, and BMI were collected to determine their potential influence on the severity of the deformity.

2.1. STATISTICAL ANALYSIS

To determine the relationship between quantitative anthropometric variables (radiographic axis, age, and BMI), Pearson's correlation coefficient (r) was calculated, using the absolute value of the deformity (LRx) to measure severity independently of the axis direction. To compare the magnitude differences between groups (men vs. women, and patients <65 years vs. ≥65 years), the Student's t-test for independent samples was used. A specific subgroup analysis was performed for varus and valgus cases to identify distinct progression patterns. All hypothesis tests were conducted with a 95% confidence interval, considering p-values <0.05 as statistically significant. The sample distribution and dispersion were assessed by calculating the mean, standard deviation, and performing frequency analysis.

3. RESULTS

The series showed a bimodal distribution of the mechanical femorotibial axis with a mean of 6.6º (SD 12.4), exhibiting a wide data dispersion and a bias towards varus deformity, present in 81 cases (Table 1). Although the varus pattern was predominant throughout the series, the incidence was significantly higher in men (77.7% versus 54.1%) (Figure 1). The mean age was slightly lower in men than in women, although this difference was not statistically significant (Table 2). The mean degree of deformity was slightly higher in women, also without reaching statistical significance (Student's t-test, p=0.312) (95% CI: -2.62–1.10). The correlation between age and deformity was strong in men and weak in women. The mean BMI was slightly higher in men, and both genders were overweight. The values for the most extreme cases were in the varus group and were similar in both genders. A weak positive correlation was observed between age and the magnitude of the deformity (r = 0.28) with high statistical significance (p < 0.05) (95% CI: 0.116-0.424) (Figure 2). The deformity increases with age: for each year of age, the deformity increases by an average of 0.23°. When segmented by type of deformity, the relationship with age is significantly stronger in cases of varus (r = 0.42) (p < 0.001) (95% CI: 0.224-0.582) than in cases of valgus (r = 0.15) (p = 0.366) (95% CI: -0.178-0.448). As the patients' age increases, the absolute value of the varus deformity tends to be greater. Regarding sex, although women presented a higher average deformity, the difference was not statistically significant (p = 0.31). BMI analysis revealed a close relationship with the type of malalignment. While the mean BMI for the overall series was 3.09 kg/m², when stratified by groups, patients with varus deformity showed a mean BMI of 32.17 kg/m² (obesity range), while in the valgus group the mean was 29.09 kg/m² (overweight range), with a highly significant difference between the two (p < 0.001).

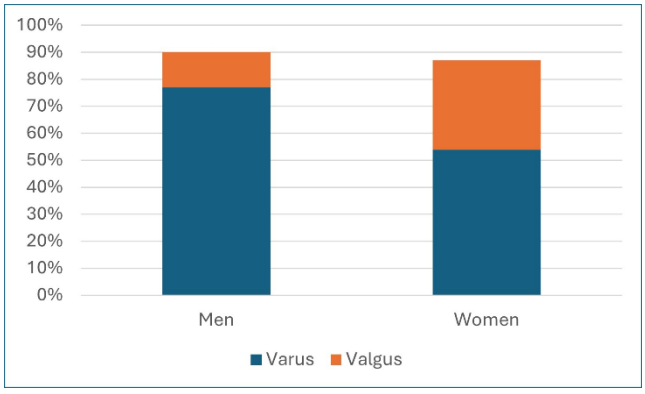


Figure 1: *The varus pattern is predominant in both genders, but with a significantly higher incidence in men.*

Table 1: Frequency of femorotibial deformity

Type	LRx*	n (%)
Valgus	< -3	34 (26.1)
Neutral	-3° a +3°	15 (11.5)
Varus	> +3°	81 (62.3)

Table 2: Gender related differences

Variable	Men (n=45)	Women (n=85)
Mean age (years)	70.4	71.3
Mean deformity	12.18°	13.56°
Age-deformity correlation	0.42	0.20
Extreme cases	+28.0°	+31.2°

3.1. SUBGROUP ANALYSIS BY AGE (<65 VS ≥65 YEARS)

When patients were divided into two age groups, older and younger than 65 years, it was observed that young men tended to have more balanced femorotibial axes, but their deformity increased dramatically with age. Young women presented with a higher initial deformity, but the increase was less pronounced with age, although they ended up with slightly higher values (Table 3).

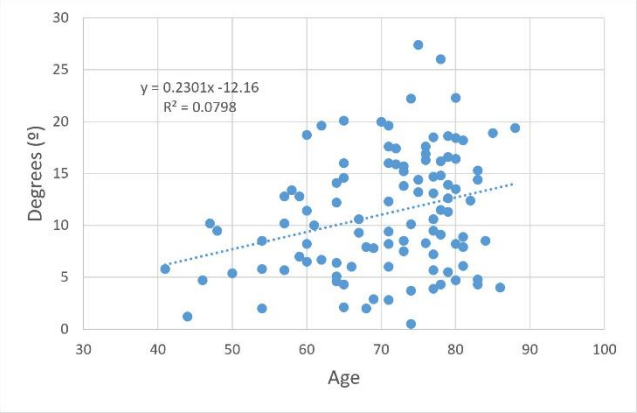


Figure 2: *Correlation between patient age and the absolute magnitude of angular deformity. The upward trend line confirms the correlation (r=0.28) (p=0.001). Cases with more than 20º of deformity are concentrated from age 70 onwards.*

Table 3: Increased deformity in relation to age

Sex	< 65 years	> 65 years	Increase
Men	8.2°	13.9°	70%
Women	11.1°	14.3°	28%

3.2. SUBGROUP ANALYSIS OF EXTREME CASES

When selecting the cases with the most severe deviations (those with more than 20° of absolute deformity, either varus or valgus) (Table 4), it was observed that 73% of the cases with extreme deformity were women. The maximum valgus reached -26° and the maximum varus +31°, and both occurred in women aged 78 years. In the 10 most severe cases of valgus, 100% were women. While extreme valgus is almost exclusively seen in older women, extreme varus affects both sexes. Regarding BMI, although it does not show a significant overall correlation, it is identified as a contributing factor in extreme deformities: patients with severe varus have a mean BMI of 31.2 (Type I Obesity), compared to 28.4 (Overweight) observed in equivalent valgus (Figure 3). When comparing the BMI of the cases with severe deformity versus the rest, a significant BMI-severe varus relationship was observed (p=0.024), which confirms that the elevation of BMI is associated with the cases with more severe varus deformity.

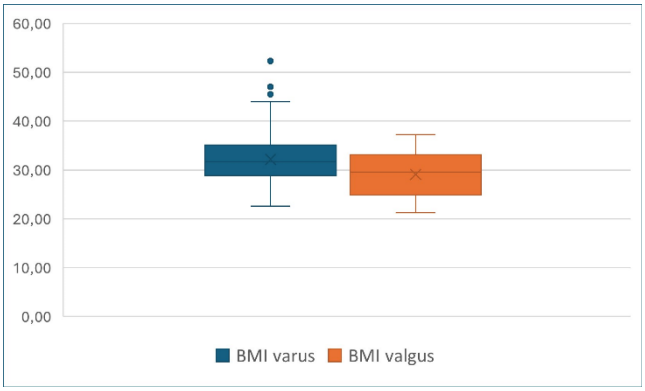


Figure 3: *BMI in extreme degrees.*

Table 4: Findings in the extreme degrees

Variable	Maximum valgus	Maximum varus
Mean age (years)	76.8	74.2
% of women	90	60
Average BMI	28.4	31.2

4. DISCUSSION

In our study, we observed that patients placed on the waiting list for TKR present specific characteristics in limb alignment, which also have a positive correlation with anthropometric factors. Varus deformity is more frequent, especially in males, and is associated with a high BMI and older age. Conversely, no relationship was found between valgus deformity and age. Women presented a greater degree of this type of deformity, such that severe valgus deformity is almost exclusive to them.

Our work was structured according to the traditional classification of the limb's mechanical axis as varus, neutral and valgus. Recently, and with the aim of facilitating the surgical technique for TKR, an attempt has been made to classify leg alignment in the coronal plane (CPAK) [2]. The nine described phenotypes, based on the constitutional axis of the knee and joint line obliqueness, have shown variations among different geographic, racial, and sexual origins [3], as well as depending on the degree of osteoarthritis [4], thus their usefulness has not been confirmed. Furthermore, it has not been possible to correlate the different phenotypes with the clinical outcomes achieved after total knee replacement [5]. To define the type of deformity and to grade its severity, we used a long-leg radiograph, including the hip, knee, and ankle. Although other techniques such as CT can also be used, the long radiograph, taken and labelled with digital techniques, is widely recommended for determining the

mechanical axis of the limb [6, 7]. More than 60% of our cases presented with varus deformity, as reported by other authors [8]. In other studies, this percentage differs, exceeding that figure [9, 10], while in others it does not reach 40% [11] or even 30% [12]. Our series presents some characteristics, such as the lower frequency of neutral cases, but this may be due to the patient selection performed in the study prior to the surgical indication. In our series, females were more frequent than males, as occurs in the presentation of knee osteoarthritis in the general population [13]. Women showed a greater tendency toward valgus, finding similar to other studies [14, 15]. Recent reports [16] have observed significant differences in the coronal axis of the limb between arthritic and healthy knees, depending on geographic or sex-related factors. The maximum values for both varus and valgus in our series were similar to those described in other articles [17]. The mean age was lower; however, the BMI was higher [15, 18], such that the average of our patients reached overweight, and the cases of varus reached grade I obesity according to the WHO scale. BMI has been associated, as in our series, with varus alignment in men but not in women [14].

We found a significant relationship between age and varus deviation. Advanced age acts as the main predictor of severity, especially in male patients with varus alignment. In men, the age-deformity relationship is more than twice as strong as in women, suggesting that age is a much more reliable predictor of increased deformity in men. This indicates a very clear linear trend: with increasing age, varus increases almost predictably. The most severe cases of varus (above 15° or 20°) are concentrated mainly in the elderly patient group (70-85 years). This finding does not coincide with that of other studies [19], which reported significant varus deviation in young patients that progressively decreased with age. The correlation found between aging and the severity of the misalignment reinforces the thesis that the deformity is not static, but rather progresses with degenerative wear, as other authors have reported [20]. While the male patient overwhelmingly tends toward varus deformity, severe valgus deformity is almost exclusively found in women, coinciding with other findings [21], which suggest specific hormonal and anatomical factors. This progression is more linear and predictable in men ($r = 0.42$) than in women ($r = 0.20$), although the latter present the most severe deformities in the series. The age-deformity relationship is clearly stronger in men: while in men the deformity appears to be a degenerative process closely linked to the passage of time, in women the deformity is more severe from younger ages and its progression is less linear.

When selecting cases with the most severe deviations, it was also observed that over 70% of cases with extreme deformity are women. In the 10 most severe cases of valgus, 100% are women. Severe valgus thus appears to be a female pathology. These findings could suggest a different surgical approach based on the patient's sex. When dividing patients into two age groups (over and under 65 years), we observed that young men tend to have much more balanced axes, but their deformity increases dramatically with age. Young women present with a higher baseline deformity, so the increase with age is less drastic in percentage terms, although they end up with slightly higher absolute values. Since we do not have references on the evolution of the deformity, we cannot discuss the findings of some authors who refer to an accelerated progression of the deformity, especially in varus [22] or that the alignment shifts slightly towards varus over time [23].

If we compare the groups according to the type of deviation, highly significant differences were found in the anthropometric profile. The group with varus deformity had a significantly higher BMI than the group with valgus deformity. This finding suggests that excessive mechanical loading is more closely associated with medial compartment collapse than with lateral compartment collapse. Varus appears to be more influenced by mechanical overload factors, while valgus, with its lower BMI may depend more on other factors (such as bone morphology or laxity) and less on excess weight. Our finding, however, is fundamental as it suggests that varus is not just a matter of age but is strongly linked to clinical obesity. While valgus could be related to structural or laxity factors, varus in our series behaves as a purely mechanical overload pathology. From a surgical perspective, the detected association between severe varus and a high BMI poses specific challenges. This suggests that, in our sample, patients who are candidates for total knee arthroplasty would require more thorough preoperative planning, anticipating more aggressive soft tissue release procedures and, possibly, the use of implants with a higher degree of constriction or stems to compensate for the poor bone stock quality caused by chronic mechanical overload. These findings are consistent with those found in extreme cases. The patient's BMI appears to act as an "accelerator" in the most severe cases. Patients with varus deformity greater than 20° have body mass indices in the obese range, indicating that excessive mechanical loading could significantly aggravate medial compartment collapse. In fact, we found several cases with BMIs above 32 and 35, reinforcing the idea that weight is an aggravating factor in varus deformity. The literature supports our findings regarding the influence of BMI on deformity. Authors such

as Felson et al. [24] have demonstrated that obesity not only initiates the degenerative process but also alters joint biomechanics. Likewise, the strong association we found between varus and a BMI greater than 31 ($p < 0.001$) coincides with the findings of Sharma et al. [25], who suggest that excessive mechanical loading in the medial compartment creates a vicious cycle of deformity progression.

If we had to define, according to our study, the patient at highest risk of severe deformity, it would be for valgus: a woman, over 75 years old, with a normal or low BMI, and for varus: a man, over 70 years old, with a BMI greater than 30.

Despite the significant findings presented, this study is not without limitations that should be considered when interpreting the results. Due to the study design, the data represent a "slice in time." Although we found a correlation between age and deformity, we cannot observe the individual progression of each patient over the years. A prospective longitudinal study would be ideal to confirm the annual rate of deformity progression. While the total sample size is sufficient for overall analysis, dividing the cases by sex or type of deformity reduces the number of subjects. This could limit the statistical power to find significant differences in specific subgroups. The study focuses on basic radiographic and age, BMI, sex. Other variables that could influence joint deformity, such as a history of trauma, were not included. As this is a case series from a specific centre and involves patients with very pronounced symptoms, there may be selection bias, which would explain the low prevalence of individuals with neutral alignment compared to the general population.

5. CONCLUSIONS

- Age-Deformity Correlation: Age is a determining factor in the progression of angular malalignment, with varus deformity showing the most pronounced progression with age. There is a positive and significant correlation between aging and the increase in knee angular deformity.
- Critical influence of BMI: BMI is significantly higher in patients with varus deformity compared to those with valgus deformity. BMI acts as a severity factor. Overweight is identified not only as a risk factor for the development of the deformity but also as an accelerator towards stages of extreme severity. Patients with more severe deformities have significantly higher BMI levels, identifying obesity as a critical risk factor for severe joint collapse.
- Sexual dimorphism and clinical patterns: There is a clear difference between genders. Varus is the predominant pattern in men, with a more predictable and linear progression with age, while women exhibit greater heterogeneity, accounting for all cases of severe valgus. This suggests distinct etiopathogenic mechanisms between the sexes.
- Surgical relevance: The combination of obesity and severe varus defines a patient profile of high surgical complexity, making weight management a primary therapeutic measure before TKR.

6. CONFLICT OF INTERESTS

The authors have no conflict of interest to declare. The authors declared that this study has received no financial support.

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